GOVT POLYTECHNIC, JAMNAGAR
DEPARTMENT OF COMPUTER ENGINEERING
SEMESTER – 5th
PROJECT REPORT GUIDELINES

CONTENTS:

Following sequence of contents is to be followed and should be strictly maintained:

1. Title Page
2. College Certificate Page
3. Acknowledgement
4. Table of Contents with Page No
5. Project Abstract
6. CHAPTERS (Main content) (***See The Chapter Details in the following pages***)

FOLLOWING FORMATTING GUIDELINES ARE TO BE STRICTLY FOLLOWED.

PAPER: Use good quality **A4 size** (210mm X 297mm) Paper.

(PRINT ON ONE SIDE OF THE PAPER ONLY and SPIRAL BIND THE WHOLE REPORT – MAKE 1 COPY PER GROUP MEMBER AND 1 COLLEGE COPY)

MARGINS :
Margins for pages including the regular text should be as below:

Left	1.2 Inches
Right	1.0 Inch
Top	1.0 Inch
Bottom	1.0 Inch

FOLLOWING MUST BE STRICTLY FOLLOWED (HEADER and FOOTER)

TOP-RIGHT CORNER	:	contains the project title
BOTTOM- RIGHT CORNER	:	contains the page number (starting from CHAP-1)

NUMBERING

- The page numbering starting from the beginning of Chapter 1 up to the end of the report should be numbered in Arabic numerals, i.e. 1, 2, 3...

PREPARATION OF CHAPTERS

Font :

Chapter Headings	**:**	Times New Roman 16 pts, **bold,** UPPER CASE
Section Headings	**:**	Times New Roman 14 pts, **bold,** UPPER CASE
Subsection Headings	**:**	Times New Roman 12 pts, **bold**, Title Case
Regular Text	**:**	Times New Roman 12 pts, normal
Special Text	**:**	Times New Roman Italics 12 pts (for foot notes, symbols, quotes..)

SPACING/ALIGNMENT

- Use 1.2 spacing between lines of regular text

- Use double spacing between Paragraphs, Chapter title and section Heading, Section heading and paragraphs, Table Title and Table, Figure Title and Figure
- Use single spacing between text corresponding to bullets, listing and quotes in the body of text.
- All paragraphs in the report are to be fully justified from the first line to the last line and should start from left without any hyphenation/indent. Text corresponding to bullets, listings and quotes in the main body of text should be indented.

SECTION/SUBSECTION NUMBERING

Section numbering should be left justified using bold print with Arabic numerals and decimals.

Example: 1.1, 2.1, 1.2.1

FIGURES, DIAGRAMS, TABLES

Figures and Tables should follow immediately after/on immediate next page after they are referred to for the first time in the text. Figure headings should be given at the bottom of the figure. All figures and tables are to be central aligned on the page. Each drawing is to be numbered and referred to as Figures only. Diagram title should be similar to figure titles. Table heading should be given at the top of the table.

MINOR PROJECT REPORT – CHAPTER STRUCTURE

1.0 INTRODUCTION

1.1 Overview of Proposed System

- Introduction to the project
- Background and motivation
- Problem being addressed
- Brief description of the proposed system
- Advantages of the proposed system

1.2 Problem Statement

Clearly define the problem that the project intends to solve.

1.3 Objectives

List the specific objectives of the project.

1.4 Scope of the Project

Describe the scope of the project.

List of Modules and Their Functions:

Module	Main Functions
Module 1	Description of functions
Module 2	Description of functions
Module 3	Description of functions

1.5 Limitations

Mention the limitations or constraints of the proposed system.

1.6 Technical Requirements and Technologies Used

Include:

- Hardware requirements
- Software requirements
- Programming languages
- Frameworks/libraries
- Database
- APIs/tools
- Development environment

2.0 SYSTEM REQUIREMENTS SPECIFICATION

2.1 User Characteristics

Describe the different types of users who will interact with the system.

Example:

- Administrator
- Registered User
- Guest User

For AI/ML projects, users may include:

- End User
- System Administrator
- Model/Developer User

2.2 Functional Requirements

Describe the functionality of each module in detail.

Example:

Module: User Registration

- User enters registration details.
- System validates the information.
- System stores the information in the database.
- System displays registration confirmation.

2.3 Non-Functional Requirements

Include:

- Performance
- Security
- Usability
- Reliability
- Availability
- Scalability
- Maintainability

Only include requirements relevant to the project.

3.0 SYSTEM ANALYSIS AND DESIGN

3.1 User-Based Modeling

3.1.1 Use Case Diagram

Provide the complete Use Case Diagram showing:

- Actors
- System
- Use cases
- Relationships

3.1.2 Use Case Description

Provide a detailed description of important use cases.

Recommended format:

Field	Description
Use Case	User Login
Actor	User
Pre-condition	User must be registered
Main Flow	Steps performed by the user
Alternative Flow	Alternative/error conditions
Post-condition	Result after completion

3.2 System Architecture

Include the overall architecture/block diagram of the proposed system.

For example:

User → Frontend/UI → Backend/API → Database

For AI/ML:

Input → Preprocessing → AI/ML Model → Prediction → Output/UI

3.3 Data Modeling

3.3.1 Data Dictionary

List the database tables and their fields.

Example:

Table	Field	Data Type	Description
User	User_ID	Integer	Unique user ID
User	Name	Varchar	User name
User	Email	Varchar	User email

3.3.2 E-R Diagram

Provide the Entity-Relationship Diagram showing:

- Entities
- Attributes
- Primary keys
- Foreign keys
- Relationships

Note: E-R Diagram and database tables are required only if the project uses a database.

4.0 USER INTERFACE DESIGN

4.1 UI/UX Design

Include screenshots/wireframes of important screens/pages.

For example:

- Login/Registration
- Home/Dashboard
- Main functionality
- Data entry screens
- Output/result screens
- Admin panel
- Reports

For each important screen, provide a short description.

4.2 Navigation / Screen Flow

Show how users navigate between major screens/pages.

Example:

Login → Dashboard → Select Module → Perform Operation → View Result

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5.0 IMPLEMENTATION

5.1 Development Environment

Describe:

- Programming language
- Framework
- IDE/editor
- Database
- Libraries/packages
- APIs
- Development tools

5.2 Module-Wise Implementation

Explain the implementation of each major module with suitable screenshots and explanations.

5.3 Important Source Code / Sample Code

Include only important/relevant code snippets.

Do not include the complete source code in the main report. The complete source code should be submitted separately.

5.4 AI/ML Model Implementation – If Applicable

For AI/ML projects, include:

- Dataset description
- Data preprocessing
- Feature extraction
- Model architecture
- Training process
- Testing process
- Hyperparameters, if relevant
- Model evaluation metrics

6.0 TESTING AND RESULTS

6.1 Testing Strategy

Describe the testing methods used.

Examples:

- Unit Testing
- Integration Testing
- Functional Testing
- System Testing
- User Acceptance Testing

Use only applicable testing methods.

6.2 Test Cases

Test Case ID	Module	Test Condition/Input	Expected Result	Actual Result	Status
TC-01	Login	Valid username/password	Login successful	Login successful	Pass
TC-02	Login	Invalid password	Error message	Error message	Pass

6.3 Results

Present the final output/results of the project using:

- Screenshots
- Tables
- Graphs/charts, where applicable
- Performance measurements

For AI/ML Projects

Include appropriate metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC-AUC, if applicable

Also mention the final model/result achieved.

6.4 Performance Evaluation

Discuss the performance of the developed system/model compared with the project objectives.

7.0 CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion

Summarize:

- Problem addressed
- Solution developed
- Major features
- Results achieved
- Objectives fulfilled

7.2 Future Enhancement

Mention possible improvements and additional features that can be implemented in the future.

8.0 REFERENCES

Include all resources used during the project, such as:

- Books
- Research papers
- Websites
- Documentation
- APIs
- Libraries/framework documentation
- Dataset sources